The Riemann Direction, Unified: Primes, Orthogonal Frames, and the Zeta Series under Machine Check (Exercise XXXVII)

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Abstract

We unify thirteen exercises of the Riemann direction into a single machine-checked development of one hundred and one declarations (eighty-nine theorems and twelve lemmas), all formalized in Lean 4 + mathlib from known mathematics alone. The development proceeds in six layers: prime structure (infinitely many primes, Fermat's little theorem, the unit groups of the prime residue rings); an observation frame built from split-complex numbers whose pseudo-norm $a^2 - b^2$ carries the two orthogonal Euler circles (radius 1 and radius i); the intersection table of the axes with the critical circle and the prime circles (a prime circle meets the critical circle iff $p \in \{2, 3\}$); the compression of the imaginary part into a coefficient axis (the product of two imaginary coefficients leaks into the real part, with sign equal to the square of the unit); the divergence–period structure of the zeta series ($\|n^s\| = n^{\mathrm{Re}\, s}$, mirror pairs collapse to 1, the functional equation pairs the divergent region with the convergent region); and the zero structure (zero reflection under the functional equation, prime-factor zeros on the imaginary axis $1 - p^{-s} = 0 \iff s \ln p = 2\pi i k$, trivial zero condition $\cos(\pi s/2) = 0 \iff s = 2k + 1$, and the critical-line observation $\mathrm{Re}\, s = 1/2 \iff \|n^s\| = n^{1/2}$). New theorems fix the tick of the real axis in the same frame: $i^i = e^{-\pi/2}$, the i-th successor of the origin i, so that the natural numbers of the real axis are iterations of i, and prime ticks are exactly the ticks that do not decompose under iteration. The Riemann hypothesis itself is not claimed; its honest boundary is stated precisely. Zero errors, zero `sorry`.

Keywords: Lean, formalization, Riemann zeta function, primes, split-complex numbers, functional equation, 0pat

1 Prime structure

The development begins with the multiplicative world of the primes. Three facts are machine-checked from mathlib: there are infinitely many primes; Fermat's little theorem holds for every prime residue field; and the unit group of the prime residue ring has order $p - 1$, making every nonzero residue a unit. These are the structural facts that the later layers of the observation frame will reuse.

Infinitely many primes. The set of primes is infinite; a machine-checked theorem of mathlib is cited directly. Every prime is positive and greater than 1, hence nonzero in every ring extension used below.

Fermat's little theorem (fermat_little). For a prime p and a residue a not divisible by p,

$$a^{p-1} \equiv 1 \pmod p.$$

In the language of the unit group: every nonzero element of $\mathbb{Z}/p\mathbb{Z}$ has order dividing $p - 1$.

The unit group (card_units_zmod_prime). The unit group of $\mathbb{Z}/p\mathbb{Z}$ has cardinality $p - 1$:

$$|(\mathbb{Z}/p\mathbb{Z})^\times| = p - 1.$$

Observable distinctness (distinct_primes_observable). Two distinct primes leave distinct residues: if $p \neq q$, then $p \not\equiv q \pmod{q'}$ for suitable modulus, so prime factors are observable in the residue rings. This is the first instance of the method used throughout: prime structure is read off from finite quotients.

2 The observation frame: split-complex numbers

The frame in which the later layers are observed is the split-complex plane: pairs (a, b) with the pseudo-norm $a^2 - b^2$. The two orthogonal Euler circles of the frame are the level sets of this pseudo-norm: the circle of radius 1 ($a^2 - b^2 = 1$) and the circle of radius i ($a^2 - b^2 = -1$). The unit $j = (0, 1)$ squares to $+1$ (split-complex), in contrast to the complex unit $i^2 = -1$. The frame carries the axes of the complex plane as its coefficient directions, and the transposition $(a, b) \mapsto (b, a)$ is an involution exchanging the two circles.

The split unit (j_sq). $j \cdot j = 1$: the unit of the second direction is a square root of $+1$, and the pseudo-norm $\|(a, b)\|^2 = a^2 - b^2$ is multiplicative.

The two circles (real_radius_circle, imaginary_radius_circle). The circle of radius 1 is $\{(a, b) : a^2 - b^2 = 1\}$ and the circle of radius i is $\{(a, b) : a^2 - b^2 = -1\}$. They are disjoint: no point has pseudo-norm both 1 and -1 (circles_disjoint).

Orthogonality (orthogonal_radii). The two radii 1 and i are orthogonal in the pseudo-metric: the circles of radius 1 and radius i are the two sheets of the same hyperbola pair, meeting at infinity. The frame is the observation system of the later sections.

Transposition (transpose_involutive, transpose_maps_one_to_i). The map $(a, b) \mapsto (b, a)$ is an involution; it exchanges the two circles (the point $(1, 0)$ maps to the point $(0, 1) = j$, the unit of the radius-i circle). The axes of the complex plane are transposed into each other, and the origin $(0, 0)$ is the unique fixed point of the involution.

3 The axes and the circles: intersections

With the frame in place, the axes of the complex plane are observed through the two circles, by solving intersection equations rather than by appealing to definitions.

The axes meet at the origin (realAxis_inter_imaginaryAxis). The real axis $\{(a, 0)\}$ and the imaginary axis $\{(0, b)\}$ intersect at the single point $(0, 0)$.

The real axis meets the unit circle (realAxis_inter_unitCircle). Solving $a^2 = 1$ on the real axis yields $\{+1, -1\}$.

The imaginary axis meets the radius-i circle (imaginaryAxis_inter_imaginaryCircle). Solving $-b^2 = -1$ on the imaginary axis yields $\{+j, -j\}$: the unit j is the intersection of the imaginary axis with the radius-i circle — its unit, not the intersection of the two axes.

The axes avoid the opposite circles (realAxis_inter_imaginaryCircle, imaginaryAxis_inter_unitCircle). The equation $a^2 = -1$ has no real solution and $-b^2 = 1$ has no real solution: the real axis does not meet the radius-i circle, and the imaginary axis does not meet the unit circle.

4 The critical line and the prime circles

The critical circle is the circle of center $(1, 0)$ and radius 1. The prime circle of a prime p is the circle of radius $\sqrt{p}$ centered at the origin. The intersection table of the axes, the critical circle, and the prime circles is machine-checked.

The imaginary axis avoids the critical circle (imagAxis_inter_criticalCircle). The point $t \cdot J$ lies on the critical circle iff $t = 0$: the imaginary axis meets the critical circle only at the origin, whereas the real axis meets it at $\{0, 2\}$.

The axes meet the prime circles (realAxis_inter_primeCircle, imagAxis_inter_primeCircle). The real axis meets the prime circle of p at $\{\pm\sqrt{p}\}$; the imaginary axis meets it at $\{\pm i\sqrt{p}\}$.

The prime circle meets the critical circle iff $p \in \{2, 3\}$ (primeCircle_inter_criticalCircle_ge5). For $p \geq 5$ the two circles are disjoint: the radius $\sqrt{p} \geq \sqrt{5}$ is too large for the disk of radius 1 around $(1, 0)$ to reach the circle of radius $\sqrt{p}$ around the origin. The circles of 2 and 3 meet the critical circle; all other prime circles avoid it. This is the first structural fact linking the primes to the critical geometry.

5 Compression of the imaginary part

The axes of the complex plane display coefficients, not units. The imaginary axis displays the real coefficient t of the imaginary part; the unit i is compressed into the axis's semantics. The consequence is observed in multiplication.

The axis shows coefficients (complex_imag_axis_coefficient). A point z of the imaginary axis satisfies $z = t \cdot i$ exactly when $z.\mathrm{re} = 0$ and $z.\mathrm{im} = t$.

The product of imaginary coefficients leaks to the real part (complex_mul_imag_leaks). The real part of $(a+bi)(c+di)$ is $ac-bd$: the product bd of the two imaginary coefficients leaves the imaginary part and appears in the real part, with the sign of the unit square $i \cdot i = -1$.

The cross terms stay imaginary (complex_mul_cross_stays_imag). The imaginary part is $ad + bc$: one imaginary coefficient times one real coefficient stays in the imaginary part; only the product of two imaginary coefficients leaks.

The split frame leaks with the opposite sign (split_mul_imag_leaks). In the split-complex frame ($j^2 = +1$) the real part of $(a + bj)(c + dj)$ is $ac + bd$: the same compression, the same leak, with sign $+1$ — the unit square again (leak_sign_is_unit_square).

6 The divergence and the period of the series

The zeta series is born as a sum of terms $1/n^s$, and each term is a product of a divergence factor and a period factor. For $s = \sigma + it$ and a positive integer n,

$$1/n^s = n^{-\sigma} \cdot e^{-it \ln n}.$$

The modulus of the term is controlled entirely by the real part; the imaginary part contributes a unit modulus and never affects convergence.

The modulus splits (zeta_term_norm_split). For $n > 0$ and $s \in \mathbb{C}$,

$$\|n^s\| = n^{\mathrm{Re}\, s}.$$

The divergence axis (the real part) controls the modulus; the period axis (the imaginary part) is a rotation of unit modulus.

Mirror pairs collapse to the unit (period_pair_reduces, divergence_pair_reduces). On the period axis, $e^{i\theta} \cdot e^{-i\theta} = 1$; on the divergence axis, $r \cdot (1/r) = 1$ $(r \neq 0)$. Both axes carry mirror pairs that annihilate to the neutral element of multiplication.

Conjugation reverses the period axis (term_conj_symmetry). For $n > 0$,

$$\overline{n^s} = n^{\bar{s}}.$$

Conjugation fixes the divergence axis (σ unchanged) and negates the period axis ($t \mapsto -t$).

The functional equation pairs the two regions (functional_equation_bridges). For $s \neq 1$ and $s \neq -n$ for all n,

$$\zeta(1-s) = 2\,(2\pi)^{-s}\,\Gamma(s)\,\cos\!\left(\frac{\pi s}{2}\right)\,\zeta(s).$$

The map $s \mapsto 1-s$ sends the convergent region ($\operatorname{Re} s > 1$) to the divergent region ($\operatorname{Re} s < 0$): the values of the divergent series are obtained from the convergent region through this symmetry — the mechanism of analytic continuation, machine-checked in mathlib (`riemannZeta_one_sub`).

The series converges exactly where the divergence factor decays (complex_zeta_convergence_region). For $\operatorname{Re} s > 1$ the zeta series converges and

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s};$$

the divergence occurs only in $\operatorname{Re} s \leq 1$, never in the imaginary direction, because $|n^{it}| = 1$ for every t.

7 The zero structure

The functional equation and the factor structure give the first machine-checked facts about the zeros.

Zero reflection (zeta_zero_reflection). If $\zeta(s) = 0$ (with s not a pole), then $\zeta(1-s) = 0$: zeros are symmetric about the critical line $\operatorname{Re} s = 1/2$. If a zero lies off the line, its reflection is a different zero (zeta_zero_offline_pair): off-line zeros come in pairs.

Prime-factor zeros lie on the imaginary axis (prime_factor_zero). For a prime p,

$$1 - p^{-s} = 0 \iff \exists k \in \mathbb{Z},\quad s \ln p = 2\pi i k.$$

Every zero of the Euler factor $1 - p^{-s}$ lies on the imaginary axis $\operatorname{Re} s = 0$: this is the precise content of the prime factorization in complex exponents. These are zeros of the factors, not of the zeta function itself; the factors' zeros are absorbed in the analytic continuation.

Trivial zeros come from the trigonometric factor (trivial_zero_condition).

$$\cos\!\left(\frac{\pi s}{2}\right) = 0 \iff \exists k \in \mathbb{Z},\quad s = 2k+1.$$

The zeros of the cosine factor of the functional equation are exactly the odd integers; the trivial zeros of the zeta function are produced by this factor.

The critical line is observable (critical_line_observation).

$$\operatorname{Re} s = \frac{1}{2} \iff \forall n > 0,\ \|n^s\| = n^{1/2}.$$

The critical line is exactly the locus where every series term has the divergence factor $n^{-1/2}$; the modulus split of the previous section makes this an iff.

8 The real axis as iteration of i

The tick of the real axis, in the observation frame of the previous sections, is not i itself; it is the i-th successor of the origin i, computed as the i-th power of i in the multiplicative group of the complex plane.

The tick is $i^i = e^{-\pi/2}$ (i_pow_i_eq_exp_neg_pi_div_two).

$$i^i = e^{-\pi/2}.$$

The value is a positive real number strictly between 0 and 1 (i_pow_i_pos_lt_one), and it is not i (i_pow_i_ne_i). The real-axis tick is the i-th successor of the origin i: the iteration count is i itself, not a natural number — this is iteration in the sense of the multiplicative group, where the i-th iterate of the generator i is the power i^i.

The natural ticks are powers of the tick (axis_tick_pow).

$$(i^i)^n = e^{-n\pi/2}.$$

The natural numbers of the real axis are iterations of i: each tick is a power of the i-th successor of the origin.

Prime ticks are the undecomposable iterations (prime_no_power_c For a prime p there are no $a, b \geq 2$ with

$$(e^{-\pi/2})^p = \left((e^{-\pi/2})^a\right)^b.$$

Iterating an iteration corresponds to multiplying the exponents ($p = a \cdot b$); the multiplicative indecomposability of a prime is exactly the power-indecomposability of its tick.

9 The completed zeta structure

The observation tool of the zero structure is the completed zeta function, which mathlib provides with its analytic properties already machine-checked.

The completed function is symmetric (xi_symmetry, xi_symmetry_zero). The completed zeta functions Λ and Λ_0 satisfy

$$\Lambda(1-s) = \Lambda(s), \qquad \Lambda_0(1-s) = \Lambda_0(s).$$

The symmetry $s \mapsto 1-s$ is the same pairing of the functional equation, now at the level of the completed functions.

The completed function is entire (xi_entire). Λ_0 is differentiable everywhere: the completed zeta function is an entire function of order one. The zeros of Λ_0 are the nontrivial zeros of the zeta function, and the symmetry plus entireness place the zero question inside the theory of entire functions — the natural next observation tool, whose honest boundary is stated in the next section.

10 Conjugation symmetry and the real-valued function Z

The functional equation pairs s with $1-s$. Conjugation pairs s with $\bar{s}$. Their interaction fixes the conjugation symmetry of the zeta function: $\overline{\zeta(s)} = \zeta(\bar{s})$ for every $s \neq 1$. The proof is machine-checked in three regions, each by a different mechanism:

$\mathrm{Re}\, s > 1$ ***(zeta_conj_of_one_lt_re).*** The Dirichlet series has real coefficients; conjugation commutes with the series termwise.

$\mathrm{Re}\, s < 0$ ***(zeta_conj_of_re_lt_zero).*** Applying conjugation to both sides of the functional equation $\zeta(1-s) = 2(2\pi)^{-s}\Gamma(s)\cos(\pi s/2)\,\zeta(s)$, the conjugate of the multiplier equals the multiplier at $\bar{s}$ (conjugation of the power, sine and Gamma factors), and the common factor cancels.

$0 < \mathrm{Re}\, s < 1$ ***(zeta_conj_of_critical_strip).*** In the critical strip neither the series nor the functional equation applies directly. The zeta function is expressed through the completed function: $\zeta(s) = (s\Lambda_0(s)-1-s/(1-s))/(2\pi^{-s/2}\Gamma(s/2+1))$, where Λ_0 is the entire completed zeta function of mathlib, defined as a Mellin transform of a real-valued kernel. Conjugation commutes with integration (Bochner integral, unconditionally), the kernel is real-valued, and the conjugate of t^{s-1} for real $t > 0$ is $t^{\bar{s}-1}$; every factor of the formula conjugates in place, and no analytic continuation argument is needed.

The function Z is real-valued (hardyZ_real). On the critical line $\mathrm{Re}\, s = 1/2$ write the functional-equation multiplier $\chi(s) = 2^s\pi^{s-1}\sin(\pi s/2)\Gamma(1-s)$, so that $\zeta(s) = \chi(s)\zeta(1-s)$. Two classical facts are machine-checked: the multiplier is an involution, $\chi(s)\chi(1-s) = 1$ (Euler's reflection formula for Gamma and the double-angle identity for sine, for non-integer s); and $\overline{\chi(s)} = \chi(\bar{s})$ (conjugation of the factors). On the critical line $\bar{s} = 1-s$, hence $\chi(1/2+it)$ has unit modulus: $\overline{\chi(1/2+it)} = \chi(1/2-it) = \chi(1/2+it)^{-1}$. The function

$$Z(t) = \chi(1/2+it)^{-1/2}\,\zeta(1/2+it)$$

is therefore real-valued: conjugation of Z uses the conjugation symmetry of ζ, the unit modulus of the multiplier, the conjugation law for the power $-1/2$ (main branch; at the branch point $\chi = -1$ the value is computed directly), and the power-law identity $\chi^{1/2}\chi^{-1} = \chi^{-1/2}$. The equality $\overline{Z(t)} = Z(t)$ is machine-checked for every real t.

11 Honest boundary and position

The content is classical mathematics: prime structure, split-complex geometry, intersection tables, the compression of the imaginary part, the divergence–period split of the series, the functional equation, zero reflection, prime-factor zeros, trivial zeros, the critical-line observation, the iteration of i, and the completed zeta structure. All one hundred and one declarations (eighty-nine theorems and twelve lemmas) are machine-checked in Lean 4 + mathlib from known mathematics alone, with zero errors and zero `sorry`; the functional equation and the completed zeta facts are cited from mathlib rather than proved anew.

The honest boundary is stated precisely: the Riemann hypothesis — every nontrivial zero of the zeta function lies on the critical line $\mathrm{Re}\, s = 1/2$ — is not claimed. What is machine-checked here is the structure around it: the symmetry of zeros under the functional equation, the location of the Euler-factor zeros on the imaginary axis, the trigonometric origin of the trivial zeros, and the observation that the critical line is exactly the locus of the divergence factor $n^{-1/2}$, and the machine-checked fact that the function $Z(t) = \chi(1/2+it)^{-1/2}\zeta(1/2+it)$ is real-valued. The Riemann hypothesis is equivalent to all zeros of Z being real; that statement is not claimed. Proving that Z has no non-real zeros requires the analytic theory of zero counting and of sign changes, which is beyond the machine-checked content of this exercise. The Riemann hypothesis remains open.

Artifact

The artifact contains `proof.lean` (Lean 4.32.2, mathlib v4.32.2): one hundred and one declarations (eighty-nine theorems and twelve lemmas), zero errors, zero `sorry`; and `observation.md`, a record of twenty numerical observations (critical strip, zero locations, base-point shift, conjugation records, and exhaustive radius searches). Build: `lean proof.lean`.